

Section I**10 marks****Attempt Questions 1-10****Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1-10.

1. Which of the following is equal to $\frac{x^2 - 36}{x - 6}$?

- (A) $x - 6$
- (B) $x + 6$
- (C) $x - 3$
- (D) $x + 3$.

2. What are the solutions to $3x^2 - 7x - 1 = 0$?

- (A) $x = \frac{-7 \pm \sqrt{37}}{6}$
- (B) $x = \frac{-7 \pm \sqrt{61}}{6}$
- (C) $x = \frac{7 \pm \sqrt{37}}{6}$
- (D) $x = \frac{7 \pm \sqrt{61}}{6}$.

3. What are the exact solutions of $2 \cos x = -\sqrt{3}$ for $0 \leq x \leq 2\pi$?

- (A) $\frac{\pi}{6}$ and $\frac{11\pi}{6}$
- (B) $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$
- (C) $\frac{\pi}{3}$ and $\frac{5\pi}{3}$
- (D) $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$.

4. Which of the following define the domain and range of the function $f(x) = \log_e x$?

- (A) Domain: all real x and Range: all real y .
- (B) Domain: $x > 0$ and Range: $y > 0$
- (C) Domain: all real x and Range: $y > 0$
- (D) Domain: $x > 0$ and Range: all real y .

5. What is the derivative of $(e^{3x} + 1)^{-2}$?

- (A) $-2e^{3x}(e^{3x} + 1)^{-3}$
- (B) $-2e^{3x}(e^{3x} + 1)^{-1}$
- (C) $-6e^{3x}(e^{3x} + 1)^{-3}$
- (D) $-6e^{3x}(e^{3x} + 1)^{-1}$.

6. What is the perpendicular distance of the point (4,5) from the line $3x - 2y + 10 = 0$?

- (A) $\frac{12}{\sqrt{13}}$
- (B) $\frac{17}{\sqrt{13}}$
- (C) $\frac{2}{\sqrt{5}}$
- (D) $\frac{12}{\sqrt{41}}$.

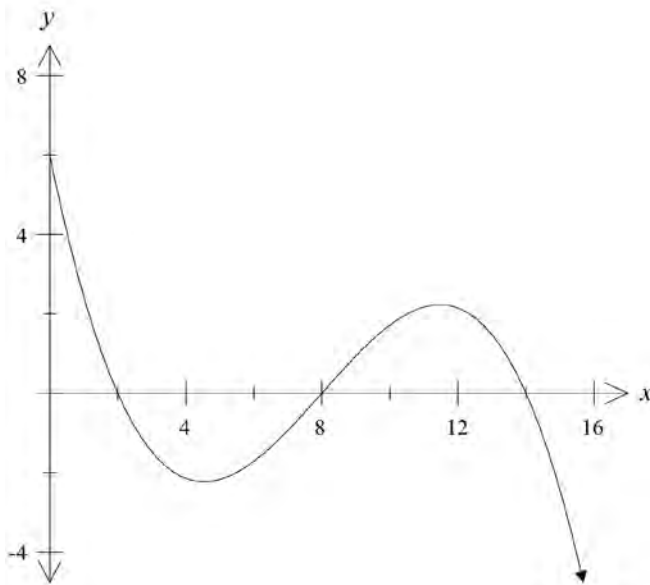
7. What is the solution of $5^x = 20$?

- (A) $\log_4 5$
- (B) $\log_5 4$
- (C) $1 + \log_4 5$
- (D) $1 + \log_5 4$.

8. A parabola has a focus (3,1) and directrix $x = 5$. What is the equation of the parabola?

- (A) $(y-1)^2 = -4(x-4)$
- (B) $(y-1)^2 = 8(x-3)$
- (C) $(x-3)^2 = -8(y-3)$
- (D) $(x-3)^2 = -16(y-1)$.

9. The diagram below shows the graph $y = f(x)$.



Where is the function increasing, at a decreasing rate?

- (A) (2,0)
- (B) (6,-1.8)
- (C) (10,1.8)
- (D) (14,0).

10. What is the value of $\int_3^5 4x^3 dx$

- (A) 192
- (B) 408
- (C) 544
- (D) 706.

Section II

90 marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Question 11-16, your responses should include relevant mathematical reasoning and /or calculations.

Question 11 (15 marks)

- a) Evaluate $\frac{\ln 5}{3}$ correct to three significant figures. 1
- b) Evaluate $\lim_{x \rightarrow 3} \frac{x^3 - 3x^2}{x - 3}$ 2
- c) Differentiate $(1 + \tan x)^4$ 2
- d) Differentiate $x \ln x$ 2
- e) Find $\int 4xe^{x^2+1} dx$ 2
- f) Evaluate $\int_0^2 \frac{3x}{x^2+1} dx$ 3
- g) Sketch the region defined by $x^2 + (y-1)^2 \geq 9$ 3

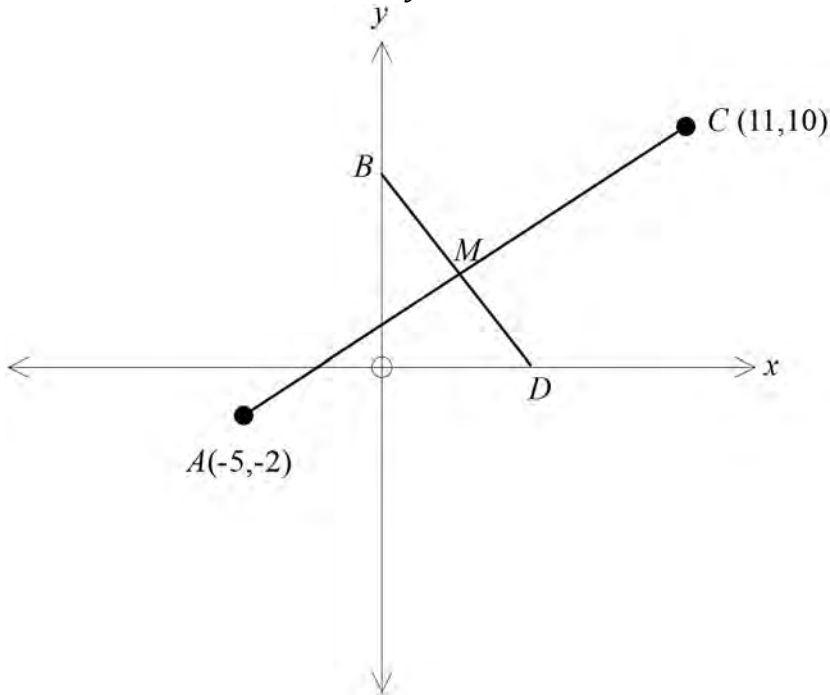
Question 12 (15 marks) Start a new booklet.

a) Given that $\int_0^4 kx + 2dx = 12$, and k is a constant, find the value of k .

2

b) $A(-5, -2)$ and $C(11, 10)$ are two points on the number plane.

M is the midpoint of AC and the perpendicular bisector of AC meets the x axis at D and the y axis at B .



i. Find the coordinates of M .

1

ii. Show that the equation of the perpendicular bisector of AC ,
i.e. line BMD , is $4x + 3y - 24 = 0$

2

iii. Hence find the coordinates of the points B and D .

2

iv. Show that the quadrilateral $ABCD$ is a rhombus.

2

c) Chairs are arranged in rows in front of a stage in a concert hall, so the row closest to the stage is the first row. Each row has two more chairs than the row in front of it. There are forty-two chairs in the tenth row.

i. How many chairs are in the first row?

2

ii. The seating arrangement has a total of 680 chairs.
How many rows of chairs are in the concert hall?

3

iii. How many chairs are in the last row?

1

Question 13 (15 marks) Start a new booklet.

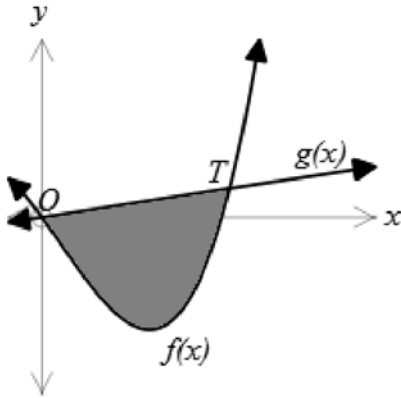
- a) The population $P(t)$ of turtles in a conservation park is given by:

$$P(t) = 200 - 75 \sin\left(\frac{\pi t}{3}\right).$$

where t is time in months.

- i. Find all times during the first 12 months when the population equals 275 turtles. 2
- ii. Sketch the graph of $P(t)$ for $0 \leq t \leq 12$. 2

- b) The diagram shows the graphs of the function $g(x) = 3x$ and $f(x) = 5x^3 - 5x^2 - 27x$.
The graphs meet at O and T .



- i. Find the x -coordinate of T . 1
 - ii. Find the area of the shaded region between the graphs of the functions. 3
- c) Tina borrows \$5000 at 1.5% per month reducible interest and pays the loan off in equal monthly instalments. Tina is to repay the loan in 3 years. 3
Calculate the value of each monthly instalment.

- d) The diagram shows two quadrants, centre O .
 $OA = 3$ cm, $OD = 2$ cm, $\angle AOP = \theta$ radians.

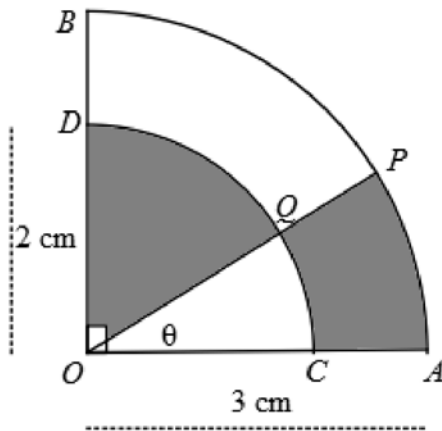


Diagram is NOT drawn to scale.

- i. Show that $\frac{5}{2}\theta$ is an expression for the area of the shaded region $APQC$. 1
- ii. If the area of the shaded region $APQC$ is $\frac{5\pi}{6}$ squared centimetres.
 Find the size of $\angle AOP$. 1
- iii. Hence find the exact area of shaded sector OQD . 2

Question 14 (15 marks) Start a new booklet.

- a) A particle travels so that its displacement (
- x
- metres), after
- t
- seconds is given by:

$$x = 12t - 3t^2.$$

- | | | |
|------|--|----------|
| i. | Where is the particle 3 seconds after it starts? | 1 |
| ii. | When does the particle turn around? | 1 |
| iii. | How far does the particle travel during the first 5 seconds? | 2 |
| iv. | Find the greatest speed during the first 5 seconds. | 1 |

- b) A cylinder is to be cut from a solid sphere.

The diagram below shows a cross section of the sphere and cylinder.

The sphere has a diameter of 8 cm. The cylinder has a height of h cm and a radius of r cm.

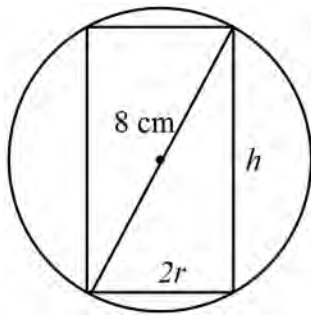


Diagram is NOT drawn to scale.

- | | | |
|-----|--|----------|
| i. | Show that the volume (V) of the cylinder is given by: | 2 |
| | $V = \pi \left(\frac{64 - h^2}{4} \right) h$ | |
| ii. | Find the value of h such that the volume of the cylinder is a maximum. | 3 |

- c) On an island, the population
- P
- after
- t
- years is given by:
- $P = P_0 e^{kt}$
- .

The initial population of the island is halved in 25 years.

- | | | |
|------|--|----------|
| i. | Show that $k = \frac{\ln 0.5}{25}$ | 1 |
| ii. | How long will it take for the population to reduce from 5000 people to 2000 people ? | 2 |
| iii. | What percentage of the original population will be present after 75 years? | 2 |

Question 15 (15 marks) Start a new booklet.

- a) i. Copy this table and complete it, leave your answers as fractions.

1

x	1	2	3	4	5
$\frac{2}{x(x+1)}$					

- ii. Use the 5 functional values from part i, and Simpson's rule, to find an approximation to

2

$$\int_1^5 \frac{2}{x(x+1)} dx.$$

Write your approximation using two decimal places.

- iii. Show that
- $\frac{2}{x} - \frac{2}{x+1} = \frac{2}{x(x+1)}$
- .

2

- iv. Deduce the value of the integral in part ii, correct to two decimal places.

2

- b) In the diagram,
- $ABCD$
- is a quadrilateral and
- BD
- is a diagonal.

$CB = 8$ cm, $AB = 9$ cm, $AD = 6$ cm and $BD = 12$ cm. $\angle DAB = \angle CBD$.

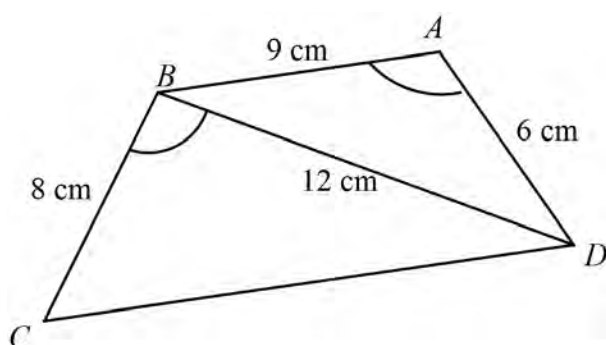


Diagram is NOT drawn to scale.

- i. Prove triangle
- ABD
- and
- BDC
- are similar.

2

- ii. Find the length of
- CD
- .

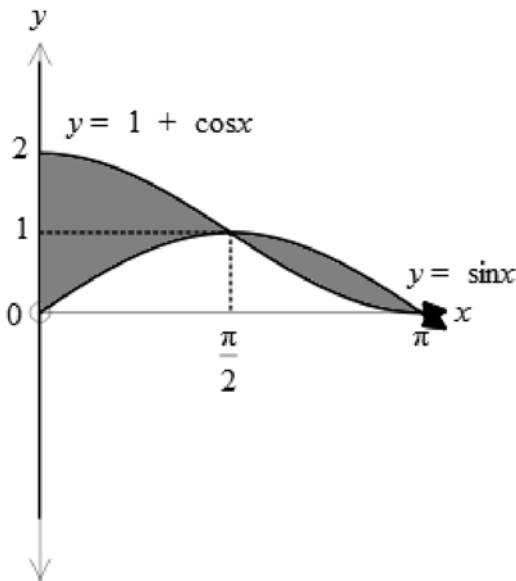
2

- iii. Prove that
- AB
- and
- CD
- are parallel

1

- c) The graphs of $y = \sin x$ and $y = 1 + \cos x$ are shown intersecting at $x = \frac{\pi}{2}$ and $x = \pi$ 3

Calculate the total area of the two shaded regions.



Question 16 (15 marks) Start a new booklet.

- a) The region bounded by the curve $y = \sec x$, the lines $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$ is rotated through one complete revolution about the x axis. 3

Find the volume of the solid of revolution. Give your answer in exact form.

- b) The acceleration of a particle is given by:

$$\ddot{x} = -12e^{-2t}$$

where x is displacement in metres and t is time in seconds.

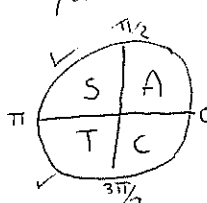
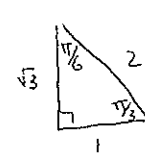
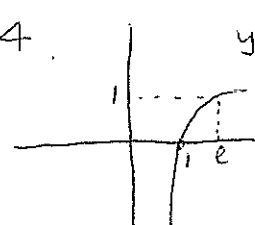
Initially its velocity is 7 ms^{-1} and its displacement is 4 m.

- i. Show that the velocity of the particle is given by: $\dot{x} = 6e^{-2t} + 1$ 2
- ii. Graph the velocity with respect to time. 2
- iii. Find the displacement when $t = 3$ seconds. 2

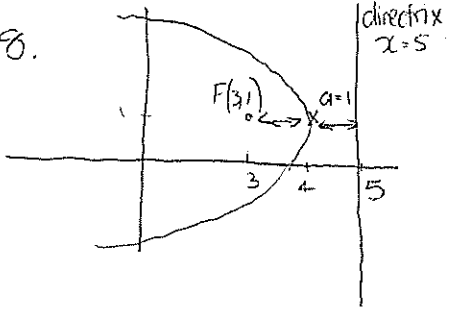
- c) Consider the function $y = 1 + 3x - x^3$, for $-2 \leq x \leq 3$.
- i. Find all stationary points and determine their nature. **3**
 - ii. Find the point of inflexion. **1**
 - iii. Sketch the curve for $-2 \leq x \leq 3$. Do not find the x - intercepts. **1**
 - iv. What is the minimum value for the curve over the stated domain? **1**

END OF THE EXAMINATION

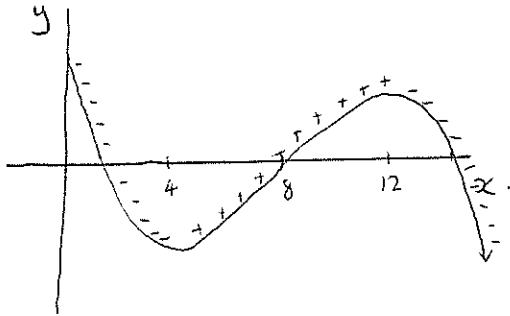
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
<u>Multiple Choice</u> Q1 Factorise, then simplify $\frac{(x+6)(x-6)}{(x-6)} = x+6 \quad \textcircled{B}$	
Q2. Quadratic Formula. $a=3 \quad b=-7 \quad c=-1$ $x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(-1)}}{2(3)}$ $x = \frac{7 \pm \sqrt{49+12}}{6} = \frac{7 \pm \sqrt{61}}{6} \quad \textcircled{D}$	
Q3. Solve, using the 4 quadrants. $2\cos x = -\sqrt{3}$ $\cos x = -\frac{\sqrt{3}}{2}$ $x = \pi - \pi/6$ and $\pi + \pi/6$ $x = \frac{5\pi}{6}$ and $\frac{7\pi}{6}$  2nd + 3rd Quadrants  \textcircled{B}	
Q4. $y = f(x) = \log_e x$  domain: $x > 0$ range: all real y \textcircled{D}	

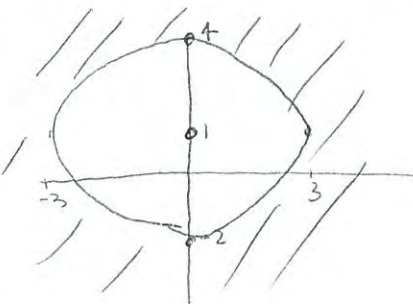
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q5. Function of a Function - Chain Rule $y = (e^{3x} + 1)^{-2}$ $\frac{dy}{dx} = -2(e^{3x} + 1)^{-3} \times 3e^{3x}$ $\frac{dy}{dx} = -6e^{3x}(e^{3x} + 1)^{-3} \quad (C)$	
Q6. x, y $a=3$ $b=-2$ $c=10$ $(4, 5)$ $3x - 2y + 10 = 0$ $d = \frac{ ax + by + c }{\sqrt{a^2 + b^2}} = \frac{ 3(4) - 2(5) + 10 }{\sqrt{(3)^2 + (-2)^2}}$ $d = \frac{12}{\sqrt{13}} \quad (A)$	
Q7. change index form to log form $\log_5 20 = x$ $\log_5 (5 \times 4) = x$ $\log_5 5 + \log_5 4 = x$ $1 + \log_5 4 \quad (D)$	
Q8.  vertex (4, 1) $(y-h)^2 = -4a(x-k)$ $(y-1)^2 = -4(x-4) \quad (A)$	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q9.  gradient indicated by + & - signs. "approximate" due to scale. function is increasing between $x=4$ and $x=12$ rate is increasing between $x=4$ and $x=8$ rate is decreasing between $x=8$ and $x=12$ (C)	
Q10. $\int_3^5 4x^3 dx = \left[\frac{4x^4}{4} \right]_3^5$ $5^4 - 3^3 = 544$ (C)	
Q11. a) 0.536 (3.5.F) b) $\lim_{x \rightarrow 3} \frac{x^2(x-3)}{(x-3)} = \lim_{x \rightarrow 3} x^2 = 9$ (1) (1) c) $y = (1 + \tan x)^4$ $\frac{dy}{dx} = 4(1 + \tan x)^3 \times \sec^2 x$ (1) $\frac{dy}{dx} = 4\sec^2 x (1 + \tan x)^3$ (1)	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q11d) $y = x \ln x$ $\frac{dy}{dx} = (1) \ln x + x \left(\frac{1}{x} \right) \quad (1)$ $\frac{dy}{dx} = \ln x + 1 \quad (1)$ e) $\int 4x e^{x^2+1} dx = 2e^{x^2+1} + C \quad (1)$ f) $\int_0^2 \frac{3x}{x^2+1} dx = \left[\frac{3}{2} \ln(x^2+1) \right]_0^2 \quad (1)$ <i>log function</i> $\frac{3}{2} \ln 5 - \frac{3}{2} \ln 1 = \frac{3}{2} \ln 5 \quad (1)$ <i>correct substitution</i> g) circle centre $(0,1)$ radius 3 (1)  test $(0,0)$ $0^2 + 1^2 \leq 9$ $\therefore$ outside of circle required (1)	Do not write $\frac{4x e^{(x^2+1)}}{2x}$ You will be penalised in HSC. Students did not check answers by differentiating after finding integral.
Q12 a) $\left[\frac{Kx^2}{2} + 2x \right]_0^4 = 12 \quad (1)$ $(8K+8) - (0+0) = 12$ $8K = 4$ $K = \frac{1}{2} \quad (1)$	Need to distinguish constants from x variables. Constants are not changed by differentiation or integration

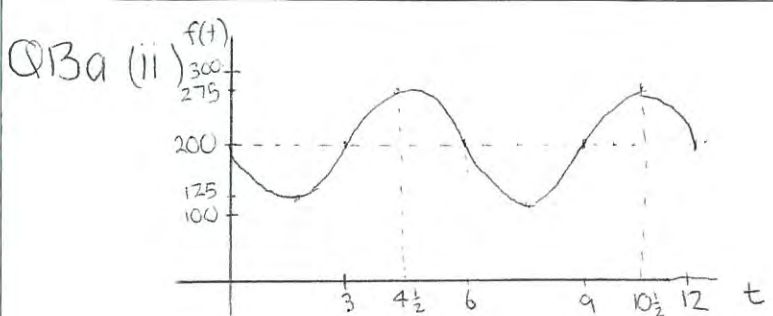
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q12 b) (i) Midpoint of AC = $\left(\frac{-5+11}{2}, \frac{-2+10}{2}\right)$ $M = (3, 4)$ ① (ii) gradient of AC = $\frac{10 - -2}{11 - -5} = \frac{12}{16} = \frac{3}{4}$ gradient of BD = $-\frac{4}{3}$ ($m_1 m_2 = -1$) ① using (3, 4) and $m = -\frac{4}{3}$ ① $y - 4 = -\frac{4}{3}(x - 3)$ $3y - 12 = -4x + 12$ ① $4x + 3y - 24 = 0$ (iii) cuts x axis when $y = 0$ $4x - 24 = 0$ $x = 6$ D (6, 0) ① cuts y axis when $x = 0$ $3y - 24 = 0$ $y = 8$ B (0, 8) ① (iv) midpoint of BD = $\left(\frac{6+0}{2}, \frac{0+8}{2}\right)$ $(3, 4)$ ① Since diagonals bisect each other at 90°, ABCD is a rhombus ①	Some students calculated the sides to be equal.

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u>	<u>comments</u>
Q12 c) $a, a+2, a+4, \dots$ A.P (i) $T_n = a + (n-1)d$ $42 = a + 9(2)$ ① $a = 42 - 18 = 24$ chairs ① (ii) $S_n = 680$ $S_n = \frac{n}{2} (2a + (n-1)d)$ $680 = \frac{n}{2} (2(24) + (n-1)2)$ ① $680 = n(24 + n - 1)$ $680 = n(n + 23)$ $n^2 + 23n - 680 = 0$ ① $(n + 40)(n - 17) = 0$ $n \neq -40$ $n = 17$ rows. ① (iii) $T_{17} = 24 + 16(2)$ $T_{17} = 24 + 32 = 56$ chairs ①	Well done. Some students did not get whole number answers. This indicates an error. 10
Q13 a) $P(t) = 275$ $200 - 75 \sin\left(\frac{\pi}{3}t\right) = 275$ $\sin\left(\frac{\pi}{3}t\right) = -1$ ① $\frac{\pi}{3}t = \frac{3\pi}{2}, \frac{7\pi}{2}, \dots$ $t = 9/2, 21/2$ ① 4.5 and 10.5 months.	Many students could not solve this.

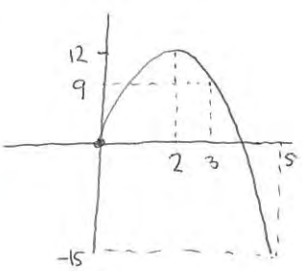
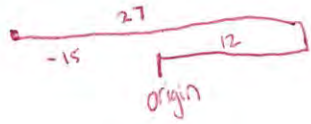
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q13a (ii)  (2) b) (i) $g(x) = 3x$ $f(x) = 5x^3 - 5x^2 - 27x$ $g(x) = f(x)$ at point of intersection T $5x^3 - 5x^2 - 27x = 3x$ $5x^3 - 5x^2 - 30x = 0$ $5x(x^2 - x - 6) = 0$ $5x(x - 3)(x + 2) = 0$ $x = 3$ $x \neq -2$ in domain shown. When $x = 3$ $y = 9$ $(3, 9)$ (1) (ii) $g(x)$ is above $f(x)$ $\int_0^3 3x - 5x^3 + 5x^2 + 27x \, dx$ $\int_0^3 -5x^3 + 5x^2 + 30x \, dx \quad (1)$ $\left[-\frac{5x^4}{4} + \frac{5x^3}{3} + \frac{30x^2}{2} \right]_0^3 \quad (1)$	Curve oscillates about 200. Need labels and scales The curve underneath is subtracted in the integral.
$\left(-\frac{5(3)^4}{4} + \frac{5(3)^3}{3} + \frac{30(3)^2}{2} \right) - 0 = 78.75 \text{ squared units}$ (1)	

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q13 c) 1st month = $5000(1.015) - R$ 2nd month = $5000(1.015)^2 - R(1+1.015)$ 3rd month = $5000(1.015)^3 - R(1+1.015+1.015^2)$ $\vdots$ 36th month = $5000(1.015)^{36} - R(1+1.015+1.015^2+\dots+1.015^{35})$ After 36th month loan has been repaid $\therefore R = \frac{5000(1.015)^{36}}{(1+1.015+1.015^2+\dots+1.015^{35})}$ $R = \\$180.76$ (1 mark) d) (i) shaded region APQC = Sector OPA - sector OQC = $\frac{1}{2}(3)^2\theta - \frac{1}{2}(2)^2\theta$ = $\frac{1}{2}5\theta = \frac{5}{2}\theta$ (1) (ii) $\frac{5}{2}\theta = \frac{5\pi}{6}$ $\theta = \frac{5\pi}{6} \div \frac{5}{2} = \frac{\pi}{3}$ (1) (iii) $\angle DOQ = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$ (1) Sector DOQ = $\frac{1}{2} \times 2^2 \times \frac{\pi}{6} = \frac{\pi}{3} \text{ cm}^2$ (1)	See alternative method in Cambridge text Missing steps GP $\leftarrow \frac{1.015^{36} - 1}{1.015 - 1} = \frac{1.015^{36} - 1}{.015}$ (1 mark) = $\frac{5000(1.015)^{36} \times .015}{(1.015^{36} - 1)}$

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
Q14 a) (i) $x = 12(3) - 3(3)^2$ $x = 9$ metres from the origin ① (ii) $\ddot{x} = 0$ whilst turning $\dot{x} = 12 - 6t$ $12 - 6t = 0$ when $t = 2$ seconds ① (iii) when $t = 0$ $x = 0$ ① when $t = 2$ $x = 12(2) - 3(2)^2 = 12$ m when $t = 5$ $x = 12(5) - 3(5)^2 = -15$ m total distance = $12 + 27 = 39$ m ①   (iv) greatest speed occurs when the gradient is steepest. test $\dot{x}$ when $t = 0$ $\dot{x} = 6 \text{ ms}^{-1}$ test $\dot{x}$ when $t = 5$ $\dot{x} = 12 - 6(5) = -18 \text{ ms}^{-1}$ greatest speed occurs when $t = 5$ ①	• very few people graphed the displacement • people who did graph managed more correct responses

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments								
Q14 b. (i) Using Pythagoras' theorem and the diagram provided $(2r)^2 + h^2 = 8^2$ $(2r)^2 = 8^2 - h^2$ $4r^2 = 64 - h^2$ $r^2 = \frac{64 - h^2}{4} \quad \text{①}$ Given Volume of cylinder $V = \pi r^2 h$ $V = \pi \left(\frac{64 - h^2}{4} \right) h. \quad \text{①}$ $V = \frac{16\pi h}{4} - \frac{\pi}{4} h^3$ (ii) max occurs when $V' = 0$ and $V'' < 0$ $V' = 16\pi - \frac{3\pi}{4} h^2 \quad \text{①}$ $V' = 0 \text{ when } h = \sqrt{16\pi \div \frac{3\pi}{4}}$ $h = \sqrt{\frac{64}{3}} = \frac{8}{\sqrt{3}}$ $\approx 4.6\text{cm}$ $V'' = -\frac{6\pi}{4} h < 0 \therefore \text{maximum} \quad \text{①}$ Volume occurs when $h = \frac{8}{\sqrt{3}} \text{ cm} \quad \text{①}$	Many students missed the Pythagoras Application. Some used incorrect formula here test for maximum Volume is required. either $V'' < 0$ or <table><tr><td>h</td><td>4</td><td>4.6</td><td>5</td></tr><tr><td>V'</td><td>+</td><td>0</td><td>-</td></tr></table> 4.62cm also accepted.	h	4	4.6	5	V'	+	0	-
h	4	4.6	5						
V'	+	0	-						

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u>	<u>comments</u>
Q14 c (i) $P_{25} = \frac{1}{2} P_0$ $\frac{1}{2} P_0 = P_0 e^{k25}$ $\frac{1}{2} = e^{k25}$ $\ln \frac{1}{2} = k25$ $k = \frac{\ln(0.5)}{25}$ ① (ii) $2000 = 5000 e^{\frac{\ln 0.5}{25} t}$ ① $0.4 = e^{\frac{\ln 0.5}{25} t}$ $\ln(0.4) = \frac{\ln(0.5)}{25} t$ $t = \ln(0.4) \div \frac{\ln(0.5)}{25}$ $t = 33 \text{ years}$ ① (iii) $P = 5000 e^{\frac{\ln 0.5}{25} \times 75}$ $P = 5000 e^{3 \ln 0.5}$ $P = 625$ ① $\% \text{ of original} = \frac{625}{5000} \times 100$ $\% \text{ of original} = 12\frac{1}{2}\%$ ①	Rounding wasn't penalised.

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Q15 a(i) $x \mid 1 \mid 2 \mid 3 \mid 4 \mid 5$ $\frac{2}{x(x+1)} \mid 1 \mid \frac{1}{3} \mid \frac{1}{6} \mid \frac{1}{10} \mid \frac{1}{15}$ ① (ii) $A \doteq \frac{1}{3} \left(1 + \frac{4}{3} + \frac{1}{6} \right) + \frac{1}{3} \left(\frac{1}{6} + \frac{4}{10} + \frac{1}{15} \right)$ ① $A \doteq \frac{47}{45} = 1.04 \quad (2dp)$ ① (iii) LHS = $\frac{2}{x} - \frac{2}{x+1}$ $= \frac{2(x+1) - 2x}{x(x+1)}$ ① $= \frac{2}{x(x+1)}$ $= \text{RHS}$ ① (iv) $\int_1^5 \frac{2}{x(x+1)} dx = \int_1^5 \frac{2}{x} - \frac{2}{x+1} dx$ $= \left[2\ln x - 2\ln(x+1) \right]_1^5$ ① $= (2\ln 5 - 2\ln 6) - (2\ln 1 - 2\ln 2)$ $= 2\ln 5 - 2\ln 6 + 2\ln 2$ $= 1.02 \quad (2dp)$ ①	using a formula incorrectly for multiple applications.

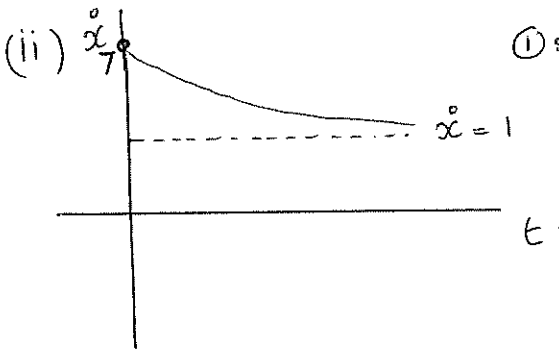
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Suggested solution(s)	comments
Q15b) (i) In $\triangle ABD$ and $\triangle BDC$ $\angle DAB = \angle CBD \text{ (given)} \textcircled{1}$ $AB : BD \quad (9:12 = 3:4)$ $AD : BC \quad (6:8 = 3:4)$ $\therefore \triangle ABD \parallel \triangle BDC$ Two sides in same proportion and included angles are equal $\textcircled{1}$ (ii) since $\triangle ABD \parallel \triangle BDC$ Then $DB : CD \quad (3:4)$ $\begin{array}{ccc} 3 & : & 4 \\ 12 & : & CD \end{array}$ $DB = 12$ $\therefore CD = \frac{12 \times 4}{3} = 16 \text{ cm} \textcircled{1}$ Corresponding sides on similar triangles have the same ratio $\textcircled{1}$ (iii) since $\triangle ABD \parallel \triangle BDC$ then $\angle ABD = \angle BDC$ alternate angles are equal therefore AB is parallel to CD $\textcircled{1}$	the abbreviations SAS is only appropriate for congruent triangles.

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Q15c) $\text{Area} = \int_0^{\pi/2} (1 + \cos x - \sin x) dx +$ $\int_{\pi/2}^{\pi} (\sin x - (1 + \cos x)) dx. \quad (1)$ $\left[x + \sin x + \cos x \right]_0^{\pi/2} + \left[-\cos x - x - \sin x \right]_{\pi/2}^{\pi} \quad (1)$ $\left[\left(\frac{\pi}{2} + \sin \frac{\pi}{2} + \cos \frac{\pi}{2} \right) - (0 + \sin 0 + \cos 0) \right] + \left[(-\cos \pi - \pi - \sin \pi) - (-\cos \frac{\pi}{2} - \frac{\pi}{2} - \sin \frac{\pi}{2}) \right]$ $\left[\left(\frac{\pi}{2} + 1 + 0 \right) - (0 + 0 + 1) \right] + \left[(1 - \pi - 0) - (0 - \frac{\pi}{2} - 1) \right]$ $= \frac{\pi}{2} + 2 - \frac{\pi}{2}$ $= 2 \text{ unit} \quad (1)$	

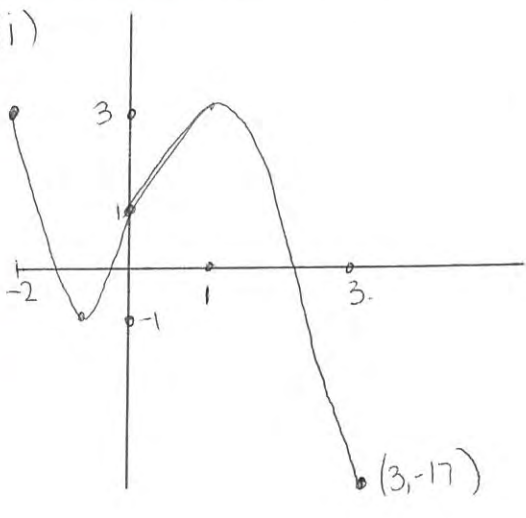
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Suggested solution(s)	comments
① 16 a) $V = \pi \int_{\pi/4}^{\pi/3} \sec^2 x \, dx$ ① $= \pi \left[\tan x \right]_{\pi/4}^{\pi/3}$ ① $= \pi \left(\tan \pi/3 - \tan \pi/4 \right)$ $= \pi (\sqrt{3} - 1)$ cubic units ① b) (i) $\ddot{x} = -12e^{-2t}$ $\dot{x} = \int -12e^{-2t} \, dt$ $\dot{x} = \frac{-12e^{-2t}}{-2} + C$ $\dot{x} = 6e^{-2t} + C$ ① when $t=0$ $\dot{x} = 7$ $\dot{x} = 6e^0 + C = 7$ $C = 1$ $\dot{x} = 6e^{-2t} + 1$ ① (ii)  ① shape (0, 7) ① asymptote.	

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Q16 b (iii) $x = \int 6e^{-2t} + 1 dt$ $x = \frac{6e^{-2t}}{-2} + t + c$ $x = -3e^{-2t} + t + c$ when $t = 0, x = 4$ $x = -3e^0 + 0 + c = 4$ $c = 7$ $x = -3e^{-2t} + t + 7 \quad (1)$ when $t = 3 \quad x = -3e^{-6} + 3 + 7$ $x = 10 - 3e^{-6} \quad (1)$									
Q16 c (i) $y = 1 + 3x - x^3$ $y' = 3 - 3x^2$ $y'' = -6x \quad (1)$ st pnt occur when $y' = 0 \quad 3 - 3x^2 = 0$ $3(1 - x^2) = 0 \quad \text{when } (1, 3) \text{ and } (-1, -1)$ when $x = 1 \quad y = 3 \quad y'' = -6 \quad (1, 3) \text{ max } (1)$ when $x = -1 \quad y = -1 \quad y'' = 6 \quad (-1, -1) \text{ min. } (1)$ (ii) $y'' = 0$ when $x = 0$ <table><tr><td>x</td><td>0^-</td><td>0</td><td>0^+</td></tr><tr><td>y''</td><td>$+$</td><td>0</td><td>$-$</td></tr></table> change of sign $\therefore (0, 1)$ is a point of inflexion. (1)	x	0^-	0	0^+	y''	$+$	0	$-$	change of sign test often omi
x	0^-	0	0^+						
y''	$+$	0	$-$						

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(iii)  when $x = -2, y = 3$ when $x = 3 y = -17$ (1) (iv) Minimum value in the domain $-2 \leq x \leq 3$ is -17 (1)	one mark was awarded for the shape of curve with tp + st pnt. students over looked the domain $-2 \leq x \leq 3$ By over looking the domain the minimum value of -17 was often missed.